AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2015

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. How is the study of the following branches of biology significant to us? (2 Marks)

i. Histology _____

_____

ii. Taxonomy _____

_____

b. State any THREE salient features of the animal shown in the diagram below. (3 Marks)



_____

_____

_____

_____

_____

_____

Q.2. (Total 4 Marks)

The data below shows the effect of different concentrations of zinc on the activity of an enzyme.

S. No.	Zinc Concentration (μg/ litre)	Rate of Enzyme Activity (Units of product produced per hour)
1	1	4
2	2	4
3	3	3.5
4	4	2

Present the given information in the form of a graph with labelled x-axis and y-axis.

Space for graph

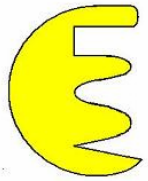
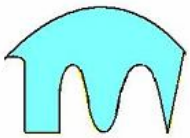
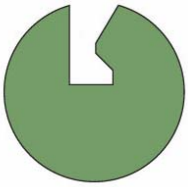
Q.3. (Total 3 Marks)

State any THREE differences between mitosis and meiosis.

S.No.	Mitosis	Meiosis
1		
2		
3		

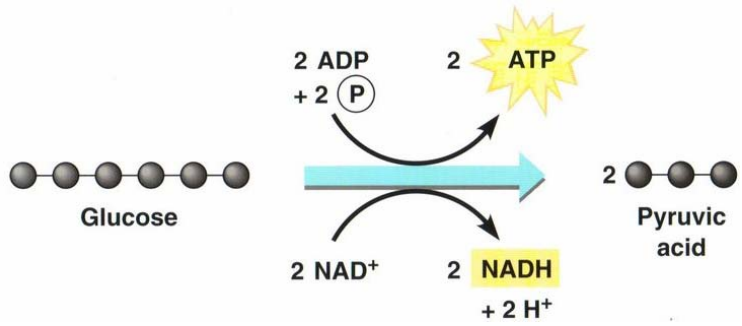
Q.4. (Total 2 Marks)

For each substrate in column II, identify and state the enzyme from column I that would act on it.

Column I Enzyme	Column II Substrate
1 	 _____
2 	 _____
3 	 _____
4 	 _____
5 	

Q.5. (Total 3 Marks)

The equation below shows part of a biochemical process.



a. Identify the process. (1 Mark)

_____

b. In which part of the cell does this process take place? Give ONE reason why it would take place in this particular part of the cell. (2 Marks)

_____

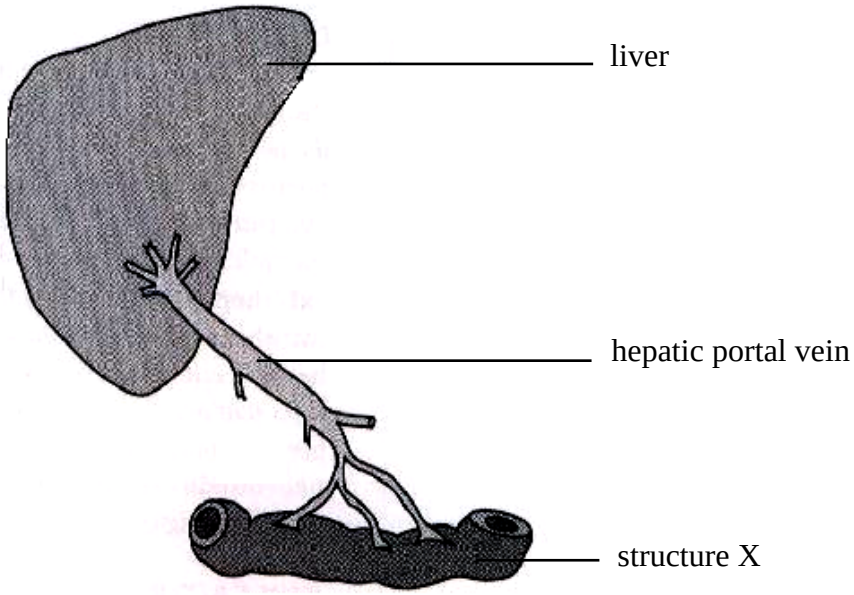
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Q.6. (Total 3 Marks)

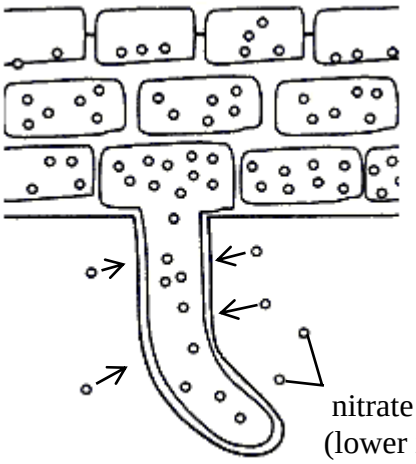
The digested food in structure X is absorbed by the blood and taken away by the hepatic portal vein to the liver.



Identify structure X in the given diagram. Also describe what happens to excess amino acids in the liver before the blood is distributed around the body.

Q.7. (Total 5 Marks)

a. The diagram below shows a root hair cell and its adjacent cells. Identify and describe the process through which plants take up the given ions from the soil. (3 Marks)



b. The pictures below show two factors contributing to famine (shortage of food). Describe any ONE of them. (2 Marks)



Picture 1



Picture 2

_____

_____

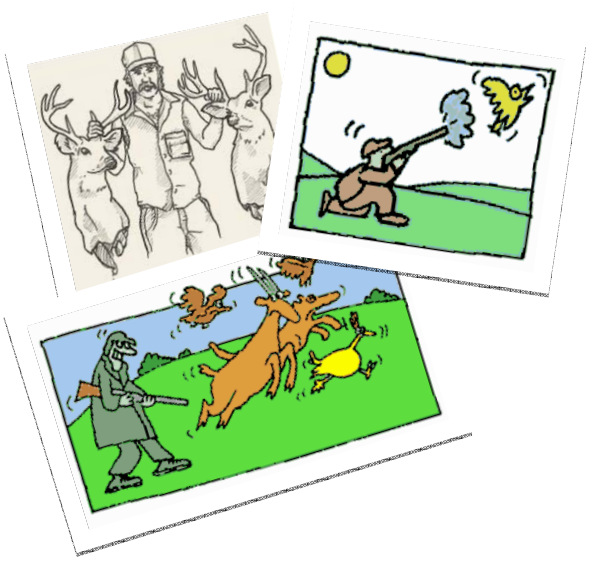
_____

_____

Q.8. (Total 7 Marks)

EITHER

a. Describe the following human activities and their impact in bringing imbalance in nature with reference to biodiversity.





OR

b. Describe the structure of endoplasmic reticulum. Also describe the structures and functions of its two types.

Q.9. (Total 8 Marks)

EITHER

a. Describe in detail FOUR protective and FOUR transport functions of blood in our body.

OR

b. A IX grade student in a 100 metre race experiences vigorous muscular contractions. Explain the biochemical process which enables him to reach the finish line. What subsequent changes will take place in his body when he takes rest after the race?

END OF PAPER